

Test 5: GUE KS test (high-T only)

Verdict: PASS | Generated: 2026-09-02 13:49:11 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

Same, restricted to high T where GUE asymptotics hold better; still data-limited (embedded 50k zeros, $T \approx 40434 \approx 10^5$).

METHOD: Same, restricted to high T where GUE asymptotics hold better; still data-limited (embedded 50k zeros, $T \approx 40434 \approx 10^5$).

Result

TEST 5 [HARDCORE pass 2]: 2 sub-checks, 0 failed $\approx$ PASS

‘plots/plot_01‘ (.png 600dpi / .svg / .pdf)

Plots

- ‘plots/animation.gif‘ $\approx$ animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v19.jl --test 5 --no-two-pass`

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

[13:49:11.305] [+ 66.587s] calc: KS D = 0.021622, p_asymptotic = 9.563953e-21 (n=49989, $\approx$ =4.8368)

[13:49:11.427] [+ 66.709s] calc: KS D = 0.021625, p_asymptotic = 9.718636e-21 (n=49958, $\approx$ =4.8360)

FULL COMPUTATION LOG

[13:49:11.431] [+ 66.713s] calc: KS D = 0.021588, p_asymptotic = 1.254559e-20 (n=49853, $\approx$ =4.8228)

[13:49:11.434] [+ 66.716s] calc: KS D = 0.021380, p_asymptotic = 4.135955e-20 (n=49523, $\approx$ =4.7605)

[13:49:11.437] [+ 66.719s] calc: KS D = 0.021203, p_asymptotic = 2.142535e-19 (n=48525, $\approx$ =4.6733)

[13:49:11.440] [+ 66.722s] calc: KS D = 0.020927, p_asymptotic = 8.613887e-18 (n=45603, $\approx$ =4.4714)

[13:49:11.443] [+ 66.725s] calc: KS D = 0.020175, p_asymptotic = 1.325911e-13 (n=37230, $\approx$ =3.8952)

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TEST 5 (secondary, HARDCORE): Objection 2: GUE Spacing Statistics $\approx$ dense high-T sweep

CONSOLE CAPTURE

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T_min	N_zeros	D	p
50	49990	0.021622	0.000000
125	49959	0.021625	0.000000
312	49854	0.021588	0.000000
781	49524	0.021380	0.000000
1953	48526	0.021203	0.000000
4883	45604	0.020927	0.000000
12207	37231	0.020175	0.000000

T2 trend : p(first)=9.564e-21 ? p(last)=1.326e-13 (must not degrade); best p=1.326e-13 (info; primary governs)

T3 top mean : mean(s) at highest band = 1.0000 (limit |?|<0.02)

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T2 trend PASS ? p 9.564e-21 ? 1.326e-13

T3 top mean PASS ? mean = 1.0000

Test 5 [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS